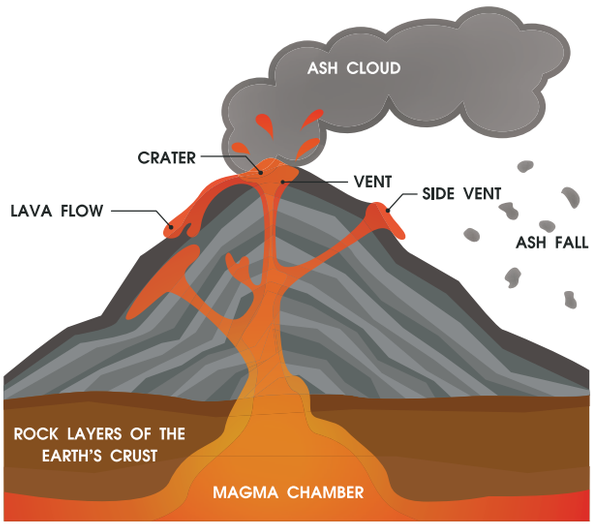


Volcano

In course of time, lava and other materials flow out of a volcano accumulate around the opening and form a conical hill or a mountain vent is an openning or mouth of a volcano. The top of this cone is usually marked by a funnel-shaped depression, which is called a *crater*. If the crater of a volcano is of great size and is shaped like a basin, it is called a *caldera*. Calderas are caused by violent explosions which blow away entire tops of great cones.

Causes of Volcanic Activity

The temperature increases as the depth increases at the rate of 1°C for every 32 metres. There is also great pressure. At a depth of about 15 km the pressure is about 5 tonnes per $cm^2$of rock. Under these circumstances, the interior of the earth is in a semi-molten state called magma. The magma, under great pressure has the capacity to dissolve great volume of gas; some gases are also combustible. This makes volcanic material burst forth through the weak spots in the earth's crust.

The scientific study of valcanoes are called **volcanology**.

People who study valcanoes are called **volcanologists**.

Nature of volcanic eruptions

Sometimes, magma rises slowly to the surface and spreads over a vast area. This is known as fissure eruption. Some plateaus and plains have been formed in this way, e.g., Deccan Plateau in India and the Colombian Plateau in North America. If the magma rises quickly to the surface, lava is thrown high into the atmosphere. Besides lava, ash, steam, gases and pieces of rocks are also thrown out. This type of eruption is known as explosive eruption. The terrible explosion on 27th August 1883 in the island of Krakatoa, Indonesia is an example for explosive type of eruption.

The viscosity of lava is determined by the amount of silica and water in magma. Highly viscosity lava is rich in silica and has little water. Low viscosity lava has little silica, but a lot of water. It moves rapidly forming smooth flows.

DO YOU KNOW?

Barren island is situated in the Andaman Sea, and lies about 138 km northeast of the territory's capital. It is only in active volcano along the chain from sumatra to myanmar. Last eruption occurred in 2017.

Types of Volcanoes

Volcanoes are classified according to their periodicity of eruptions and the state of activity such as

1. Active Valcano
2. Dormant Valcano
3. Extinct Valcano

1. Active Valcano

Valcanoes that erupt frequently are called active volcanoes. Most of the active volcanoes lie in the Pacific Ring of Fire belt which lies along the Pacific coast. There are about 600 active

7th_SS_Geography_Unit_1.indd 149 23-01-2025 12.14.41 PM